Urban green scenes: Gaps between aesthetic green view accessibility and realized flows

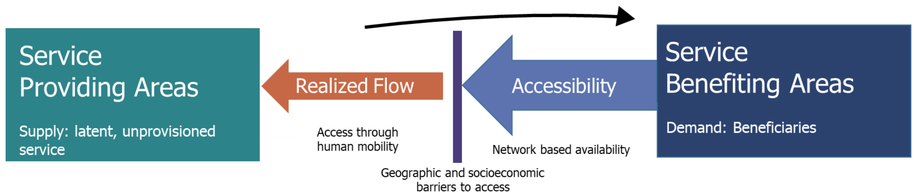
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Introduction

Aesthetic green views (AGV) can improve quality of life for human beneficiaries, with delivery dependent on both the supply and actual flows, or the use of services, by human beneficiaries (Villamanga et al., 2013; Constanza, 2008). This study explores the variable spatial patterns of each dimension and the implications of each.

Methods Overview



Ground Truth

OneFormer vegetation classes



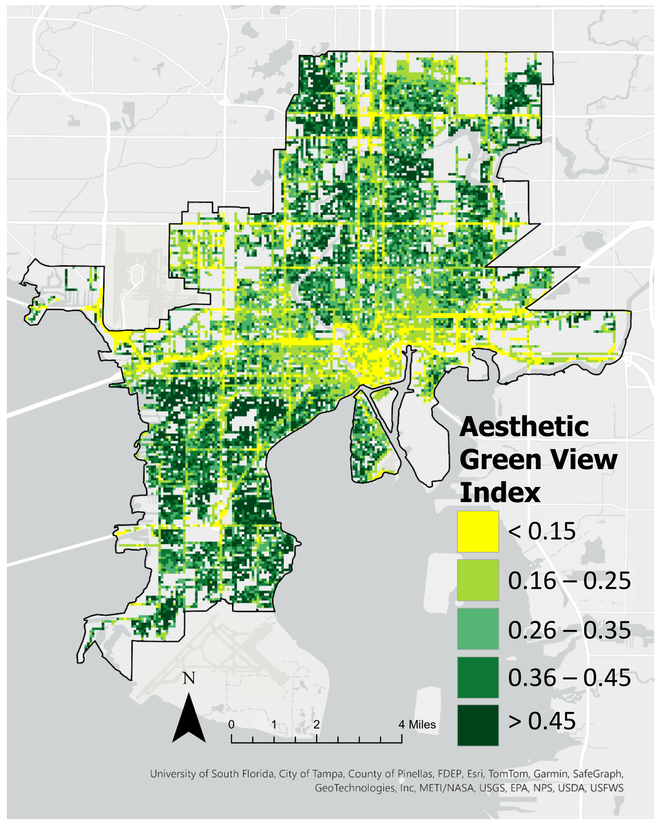
OneFormer is a transformer-based model pretrained on the ADE20K dataset. On an annotated dataset of 100 street view images, the model achieves AP = 49.4 and mIoU = 65.2 across relevant vegetation classes

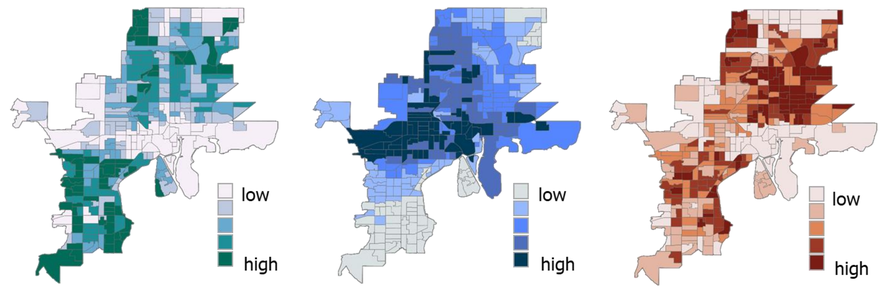
	Definition	Measurement	Source
Supply	Latent AGV, independent of human presence.	In place mean	Schirpke et al., 2019
Potential Accessibility	The network-based availability of AGV within a 30-minute drive from one's home.	Enhanced two-step floating catchment area (E2SFCA)	Ala-Hulkko, et al., 2016
Realized Flow	The actual level of access to AGV that beneficiaries achieve	AGVI of visited Points of Interest, weighted by visitation volume	Schröter et al., 2014 Wang et al., 2022

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Results

- Bivariate analysis shows that indicators of higher socioeconomic status (SES), such as income, educational attainment, home and car ownership, have a significant positive correlation with in-place supply of AGV and a negative correlation with potential flow.
- A Durbin spatial regression model (pseudo r^2 = 0.59, spatial pseudo r^2 = 0.51) shows that the interaction between in-place supply and accessibility is significant predictor of higher realized flows ($p<0.05$).







Discussion

- There are stark inequities in AGV supply, particularly in lower income, minority communities. Despite higher network accessibility to green POIs, these communities do not experience the benefits of such connectivity – indicating a disconnect between access and use.
- Communities with low supply frequently show low realized flow. However this isn't a perfect relationship, indicating spatial mismatches in AGV delivery.
- Patterns of inequity are present, but less dramatic for realized flows than home supply.

High accessibility in low SES block groups provides opportunities for improved flows through transit. Ultimately, inequities are most pronounced in AGV supply, providing strong support for greening initiatives to have a strong environmental justice component.

References